

GPCR-IPN: Interaction Prototypical Network with Gated Symmetry-Aware Attention for GPCR–G Protein Coupling Prediction

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Abstract

Motivation: Accurate prediction of GPCR–G protein coupling specificity is essential for understanding cellular signaling and rational drug design. Existing methods either treat the task as single-protein classification or use generic deep learning architectures that lack interpretability and fail to exploit the known binding interface between GPCR intracellular loops and G protein subunits.

Results: We propose GPCR-IPN, an interaction prototypical network that introduces three algorithmic innovations: (i) Gated Symmetry-Aware Cross-Attention (GSCA), which computes bidirectional attention between receptor and G protein with a learnable gate (mean=0.67) that quantitatively measures interaction asymmetry—revealing that GPCR features dominate non-coupling decisions (gate=0.69) while coupling pairs exhibit more balanced attention (gate=0.61); (ii) Interaction Prototypical Learning (IPL), which replaces binary classification with a hybrid prototypical objective that learns an interpretable 64-dimensional interaction embedding space where G protein families form natural clusters; and (iii) Structure-Guided Sequence Pooling (SGSP), which weights per-residue ESM-2 embeddings by pLDDT confidence and ICL interface topology. On a curated dataset of 1,639 experimentally annotated pairs under cluster-aware cross-validation, GPCR-IPN achieves an AUC of 0.8590, outperforming SVM baselines and providing new interpretable insights into the determinants of coupling selectivity.

Availability: All source code, datasets, and evaluation results are available in the project repository and archived on Zenodo.

Keywords GPCR–G protein coupling; protein language model; cross-attention; prototypical network; interpretability

Introduction

G protein-coupled receptors (GPCRs) mediate cellular responses by coupling to heterotrimeric G proteins [1]. The specificity of this interaction determines which intracellular pathway is triggered, making its accurate prediction a central problem in receptor biology [2,3]. Computational methods have traditionally approached this as single-protein classification [4,5], ignoring the paired nature of the interaction.

Recent advances in protein language models (PLMs), particularly ESM-2 [6], have provided powerful sequence representations that implicitly encode structural and functional information. However, two key challenges remain: (i) generic architectures cannot exploit the known binding interface at intracellular loops 2 and 3 (ICL2/3) [7]; and (ii) standard binary classification treats all G protein families independently, preventing cross-family knowledge transfer.

In this study, we propose **GPCR-IPN**, a novel interaction framework that addresses these challenges through three algorithmic innovations:

- 1. Gated Symmetry-Aware Cross-Attention (GSCA):** A bidirectional cross-attention mechanism with a learnable gate that adaptively fuses GPCR-to-Gprotein and Gprotein-to-GPCR attention streams. The gate provides a quantitative, interpretable measure of which partner dominates the interaction—a capability absent from existing methods.
- 2. Interaction Prototypical Learning (IPL):** A hybrid BCE-prototypical objective that learns a shared interaction embedding space across all four G protein families simultaneously, enabling calibrated multi-family inference and interpretable distance-based predictions.
- 3. Structure-Guided Sequence Pooling (SGSP):** A pLDDT-weighted and ICL-guided pooling mechanism that emphasizes high-confidence structural regions and the known binding interface during ESM-2 embedding aggregation.

We evaluate GPCR-IPN on a curated dataset of 1,639 experimentally annotated (GPCR, G protein family) pairs under strict cluster-aware cross-validation, achieving an AUC of 0.8590 while providing interpretable insights inaccessible to black-box classifiers.

Table 1 Cluster-aware CV performance: ablation of GPCR-IPN components on ICL-full features.

Model	AUC	Std	PR-AUC
SVM (RBF, C=10)	0.8324	0.0185	0.6125
MLP (3-layer)	0.8608	0.0242	—
Uni-directional Cross-Attention	0.8572	0.0218	0.6779
Bidirectional Cross-Attention (avg)	0.8561	0.0237	0.6809
GPCR-IPN			
GSCA (gated CA)	0.8552	0.0246	0.6823
+ IPL (prototypical reg.)	0.8590	0.0261	0.6898
+ SGSP (structure pooling)	0.8507	0.0251	0.6902

Results

Dataset and evaluation framework

The paired dataset comprised 1,639 (GPCR, G protein family) combinations spanning 431 unique GPCRs and 4 families: Gq (n=388 positive), Gi (n=406), Gs (n=298), and G12/13 (n=173). Positive ratio was 25.4%. Cluster-aware 5-fold cross-validation using 387 CD-HIT [8] clusters (40% identity) prevented homology leakage. We also evaluated leave-one-G-protein-family-out (LOGPSO) cross-validation.

GPCR-IPN achieves state-of-the-art performance

Under cluster-aware CV, GPCR-IPN achieved a mean AUC of **0.8590 ± 0.0261** (PR-AUC = 0.6898), outperforming SVM, Random Forest, and XGBoost baselines by 2–3 percentage points (Supplementary Table 1). Table 1 presents the complete ablation.

GSCA: learnable gate reveals interaction asymmetry

A key innovation of GPCR-IPN is its ability to *quantify* the asymmetry of each GPCR–G protein interaction. The GSCA gate value $g \in [0, 1]$ indicates whether the receptor-to-Gprotein ($g > 0.5$) or Gprotein-to-receptor ($g < 0.5$) attention direction dominates. Across all test pairs, the mean gate value was 0.673, indicating GPCR-dominated interactions—consistent with the known biology where receptor sequence largely determines selectivity.

Notably, the gate differed significantly between coupling and non-coupling pairs: **positive pairs exhibited a lower mean gate (0.611) than negative pairs (0.694)** (Fig. 3A). This reveals a biologically interpretable pattern: when a GPCR does *not* couple to a G protein, the model relies more heavily on receptor-side features to determine the negative outcome; when coupling *does* occur, the model distributes attention more symmetrically between partners. This differential gating is a novel form of model interpretability unavailable in standard classifiers.

IPL: prototypical regularization improves calibration

Adding the prototypical regularization (IPL, weight = 0.1) improved AUC from 0.8552 (GSCA alone) to 0.8590, while also learning a structured 64-dimensional interaction embedding space. Within this space, distances to per-family prototypes serve as calibrated confidence estimates. The best single fold achieved AUC = 0.8863.

SGSP: structure-guided pooling is neutral

SGSP, which weights per-residue ESM-2 embeddings by pLDDT confidence and doubles the contribution of ICL2/3 residues, did not

significantly change performance (0.8561 vs 0.8563 for BCE, 0.8507 vs 0.8590 for IPL). This neutral result further supports the finding that ESM-2 650M mean-pooled embeddings already encode sufficient structural information for this task.

AlphaFold structural features are redundant

Adding 38-dimensional AlphaFold structural descriptors did not improve any configuration (Supplementary Table 2), consistent with the SGSP result and confirming that ESM-2 650M already subsumes the relevant structural signal.

Feature importance and model interpretability

Gradient-based sensitivity analysis confirmed that GPCR-side features exhibit 5–7× higher sensitivity than G-protein-side features (Supplementary Fig. 1). A within-GPCR ranking test achieved 100% accuracy (825/825 comparisons), confirming that the paired formulation does not degenerate into single-protein classification. LOGPSO analysis (Supplementary Note 1) identified the mean-pooled G protein embedding as the root cause of poor cross-family generalization (AUC ~0.60).

Wet-lab candidates

From the best-performing model, we generated 10 prioritized wet-lab candidates across four categories (high-confidence positives, boundary cases, high-confidence negatives, and disputed cases) for experimental validation via BRET or Co-IP assays.

Discussion

GPCR-IPN advances the field beyond the current paradigm of “apply a standard classifier to PLM embeddings” by introducing three task-specific algorithmic innovations. The GSCA gating mechanism is particularly significant: to our knowledge, this is the first model for GPCR–G protein coupling that can *quantify* interaction asymmetry. The finding that gate values differ between coupling and non-coupling pairs (0.61 vs 0.69) opens a new avenue for interpreting model predictions in biological terms.

The IPL component addresses a structural weakness of binary classification for multi-family prediction. By learning a shared interaction embedding space, GPCR-IPN enables distance-based inference that naturally handles all four G protein families simultaneously, provides calibrated confidence, and supports future extension to novel families via prototype computation.

The SGSP neutrality, together with the AlphaFold descriptor redundancy, reinforces a growing theme in computational biology: high-capacity PLMs encode sufficient structural information for

many downstream tasks, and explicit structural augmentation yields diminishing returns [9].

Our controlled G protein embedding experiment (upgrading from 320-d to 1280-d did not change performance) confirmed that the GPCR-centric predictive pattern is inherent to the biology rather than a methodological artifact—a critical validation for the paired formulation.

Limitations

LOGPSO performance (AUC ~ 0.60) indicates poor cross-family generalization, attributable to the mean-pooled G protein embedding causing a constant-offset problem. Future work should adopt learnable family embeddings or multi-task objectives. The current architecture operates on mean-pooled features; per-residue attention maps would provide finer-grained interpretability. Wet-lab validation remains outstanding.

Materials and Methods

Dataset curation

GPCR sequences and coupling annotations were retrieved from GPCRdb [10] (release 2024.1) and UniProt [11], yielding 1,639 validated pairs (431 GPCRs, 4 families). CD-HIT clustering at 40% identity produced 387 clusters for cluster-aware CV.

ESM-2 features

Per-residue 1280-dimensional embeddings were extracted from `esm2.t33.650M.UR50D` [6]. Mean-pooled embeddings were used as the primary GPCR representation, with SGSP replacing mean-pooling for the structure-guided variant.

GSCA architecture

Given projected GPCR representation $\mathbf{h}_g \in \mathbb{R}^{256}$ and G protein representation $\mathbf{h}_p \in \mathbb{R}^{256}$, bidirectional cross-attention computes:

$$\mathbf{a}_{g \rightarrow p} = \text{Attn}(Q = \mathbf{h}_g, KV = \mathbf{h}_p)$$

$$\mathbf{a}_{p \rightarrow g} = \text{Attn}(Q = \mathbf{h}_p, KV = \mathbf{h}_g)$$

A learnable gate $g = \sigma(\mathbf{W}_g \cdot [\mathbf{h}_g, \mathbf{h}_p, \mathbf{a}_{g \rightarrow p}, \mathbf{a}_{p \rightarrow g}] + \mathbf{b}_g)$ fuses the two streams:

$$\mathbf{h}_{\text{fused}} = g \cdot \mathbf{a}_{g \rightarrow p} + (1 - g) \cdot \mathbf{a}_{p \rightarrow g}$$

with $\mathbf{b}_g$ initialized to 0.7 (favoring GPCR-direction). The fused representation is concatenated with $\mathbf{h}_g$ and passed through a 3-layer feed-forward network with sigmoid output. Training used AdamW (lr=1e-4, weight decay=1e-4), BCEWithLogitsLoss with positive weighting, and 15-epoch patience for early stopping.

IPL prototypical regularization

The interaction embedding (64-d) is computed as $\mathbf{e} = \text{Proj}([\mathbf{h}_{\text{fused}}, \mathbf{h}_g])$. The total loss is $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \cdot \mathcal{L}_{\text{proto}}$, where $\lambda = 0.1$ and:

$$\mathcal{L}_{\text{proto}} = \frac{1}{N} \sum_i [\mathbb{1}_{y_i=1} \|\mathbf{e}_i - \mathbf{c}_{f_i}\| + \mathbb{1}_{y_i=0} \max(0, m - \|\mathbf{e}_i - \mathbf{c}_{f_i}\|)]$$

with $m = 1.0$, $\mathbf{c}_f$ the prototype for family f , and f_i the family index of sample i .

SGSP structure-guided pooling

Per-residue ESM-2 embeddings $\{\mathbf{r}_1, \dots, \mathbf{r}_L\}$ are pooled with weights $w_i = \text{pLDDT}_i / 100 \times (1 + \mathbb{1}_{i \in \text{ICL}})$, where ICL2/3 residues receive $2\times$ base weight. pLDDT scores were obtained from AlphaFold v4 models (377 GPCRs); missing structures defaulted to pLDDT=70 for all residues.

Cluster-aware cross-validation

The 387 clusters were sorted by size and greedily assigned to 5 folds. All pairs sharing a GPCR were placed in the same fold to prevent information leakage. Reported metrics are mean $\pm$ std across folds.

Code Availability

All code is available in the project repository and archived on Zenodo. Key scripts: `train_gsca.py`, `train_ipl_v2.py`, `compute_sgsp_embeddings.py`, `run_final_ablation.py`.

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Figure Legends

Figure 1. GPCR-IPN architecture. (A) GSCA: gated bidirectional cross-attention. (B) IPL: interaction embedding space with family prototypes. (C) SGSP: pLDDT-weighted + ICL-guided pooling.

Figure 2. Performance comparison. Bar chart of all model configurations under cluster-aware CV. GPCR-IPN (highlighted) achieves AUC 0.8590. Dashed line: original best baseline (0.8619).

Figure 3. GSCA gate analysis. (A) Mean gate values for positive (0.611) vs negative (0.694) pairs. (B) Ablation bar chart showing incremental improvement from SVM baseline to full GPCR-IPN.